Soundness of sequent derivability

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author corollary labelled cor:entailment-soundness (2281-2283); the proof span 2285-2293 is the author proof environment that immediately follows it.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/sequent-calculus, counted once: soundness of sequent derivability. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Sequent derivability. Gamma derives A when some finite subset Gamma_0 of Gamma has a derivation of the sequent Gamma_0 with succedent A.. It produces the finite subset Gamma_0 the proof works with (2285-2288).
- Soundness of the sequent calculus. Every derivable sequent is valid; the unit proves this as its soundness theorem.. It upgrades the sequent Gamma_0 to A to its semantic validity (2288-2291).

Source reference. Original theorem thm-sequent-soundness: sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex, source lines 1982–1982. The author proof cites the unit's soundness theorem. The load-bearing work of soundness is carried by that separately proved theorem of the same unit; the counted proof retains its own argument in full.

Corollary 0.1. *If* $\Gamma \vdash !A$ *then* $\Gamma \models !A$.

Proof. If $\Gamma \vdash !A$ then for some finite subset $\Gamma_0 \subseteq \Gamma$, there is !!aderivation of $\Gamma_0 \Rightarrow !A$. By thm-sequent-soundness, every !!structure $\mathfrak{A}M$either makes some $!B \in \Gamma_0$ false or makes $!A$ true. Hence, if $M \models$ then also $M \models$. □

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